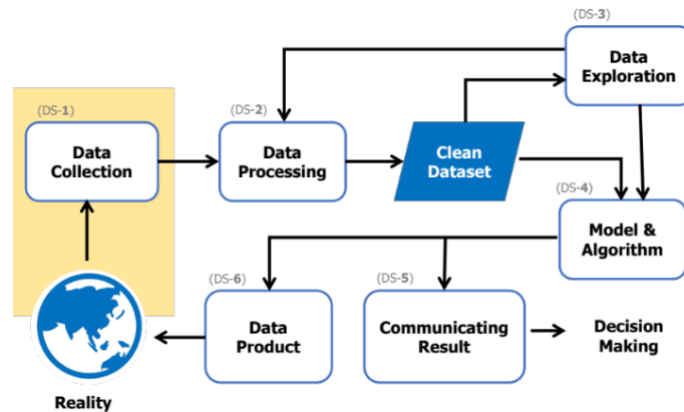


## 2. Data Collection

*“With data collection, ‘the sooner the better’  
is always the best answer.”*  
— Marissa Mayer



**Figure 2.0-1: Data collection step in the data science process**

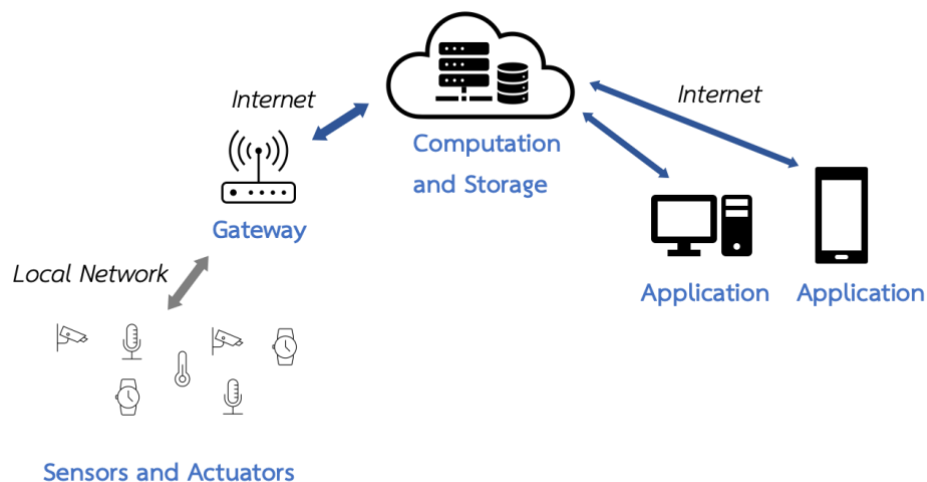
Human activity, the environment, the Earth, and the universe itself have generated enormous amounts of data since the beginning of time. Advances in digital technology have made it far more effective to sense and record data from the real world, marking the starting point of the data science process shown in Figure 2.0-1.

This chapter covers Internet of Things (IoT) technology, structured and unstructured data, data scales — which play a key role in later chapters — data collection practices, and the example datasets used throughout this book.

### 2.1 Internet of Things

The Internet of Things (IoT) is a major technological milestone (Figure 2.1-1) that lets devices capture data and be controlled over the internet, sending measured data to a server for integration, processing, and analysis — enabling many new innovations (Serpanos & Wolf, 2017).

IoT's key role is collecting data from sensors — temperature, humidity, light intensity, wind force, acceleration, speed, tilt, image, sound, heart rate, pressure, and many others — directly into digital form, without a person manually writing it down as in the past, making data integration for analysis far more efficient.



**Figure 2.1-1: General architecture of an IoT system**

## 2.2 Data Types

Electronic data can be grouped into three types: structured data, unstructured data, and semi-structured data.

### Structured Data

Structured data has a clear structure, so a specific part of it can be accessed directly by index — tabular data is the classic example.

A table (or data frame) is the most popular structure, since it is easy to understand and process. It works like a matrix: any cell is accessed by row and column index, and the table has both a row index and column headers, as shown in Table 2.2-1, which has columns Name, Age, and Height, and three rows (0, 1, 2).

When a computer reads this data and needs Billy's age, it simply looks up row 1, column “Age,” and gets the value 20 (years).

**Table 2.2-1: Example student age and height data**

|   | Name  | Age | Height |
|---|-------|-----|--------|
| 0 | Alin  | 15  | 160    |
| 1 | Billy | 20  | 180    |
| 2 | Chao  | 19  | 175    |

Structured data is commonly saved in formats such as CSV (Comma-Separated Values), TSV (Tab-Separated Values), JSON (JavaScript Object Notation), or XML (Extensible Markup Language).

This book focuses on CSV files, since they are plain text files that can easily be created with a text editor or Microsoft Excel. Each row represents one record; the first row typically holds the column names, with commas separating each field — as in the example below, representing Table 2.2-1. Because of its simplicity, standardization, and wide compatibility with many programs, CSV is extremely popular for data analytics.

```
Name, Age, Height
Alin, 15, 160
Billy, 20, 180
Chao, 25, 175
```

### Unstructured Data

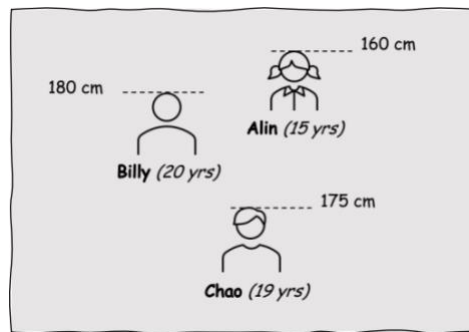
Unstructured data lacks enough structure for a computer to reliably extract specific values from it. It is usually data that humans understand directly — text, images, and sound.

#### *Text*

For example: “There are three students: Alin, Billy, and Chao. Alin is 15 years old and 160 cm tall. Billy is twenty years old and one meter eighty tall. Chao was born a year after Billy, and is half a foot taller than Alin.” A human can read this natural-language passage and work out each person's name, age, and height. But even though the text exists digitally as a string of characters, it is difficult for a computer to correctly interpret it — for instance, to extract Chao's exact age.

#### *Image*

Images (and video) are understood by humans through vision and interpretation by the brain (Figure 2.2-1) — a person can immediately read off Billy's age and height from a photo. But to a computer, an image is simply a matrix of pixel color values; it is hard for a computer to tell which part of the image is a person, which part represents height, and so on.



**Figure 2.2-1: Image example of student age and height data**

### Sound

Sound captured from a spoken sentence, e.g., reading aloud “There are three students: Alin, Billy, and Chao...” produces a waveform such as in Figure 2.2-2. Even though it's digital and a human can listen and understand it, a computer likewise finds it hard to identify which part of the waveform corresponds to a name, an age, or a height.



**Figure 2.2-2: Simulated waveform for the student age and height example**

### Semi-Structured Data

Semi-structured data has some structure but cannot be processed as conveniently as fully structured data, since most of its content is still unstructured — for example, Twitter hashtags or webpage content.

For a tweet such as “Data is the new oil. #datascience #quote #humby,” the hashtags help categorize or signal intent, but a computer still cannot easily determine what #datascience, #quote, or #humby actually mean, or who is the source of the quote, despite the #humby hashtag.

Similarly, a webpage is structured according to HTML (Hypertext Markup Language) tags, but these tags are still hard for a computer to interpret meaningfully, and different designers structure their pages differently with no single standard. The following HTML file describes a Data Analytics course, yet a computer cannot tell which part is the course title, which is the description, and which is the instructor's name:

```
<html>
<head>
<title>Data Analytics</title>
</head>
<body>
<h1>Data Analytics</h1>
<p><strong>Course Description:</strong>
This course presents a wide range of data analytic
techniques and is structured around the broad contours
of the different types of data analytics, namely,
descriptive, predictive, and prescriptive analytics.
</p>
<p><strong>Lecturer:</strong> Rathachai Chawuthai</p>
</body>
</html>
```

For this reason, semi-structured data usually needs pre-processing to convert it into a structured form before analysis, with the difficulty depending on the complexity of the underlying content.

Data analytics workflows generally favor converting datasets into tabular form first, since it has a clear structure and can be processed efficiently. Still, much of the world's data remains unstructured or semi-structured — text, images, sound, webpages — and would be extremely valuable if it could be analyzed

directly. Fortunately, ongoing advances in artificial intelligence continue to improve computers' ability to interpret such data, and it's reasonable to expect that AI will eventually understand unstructured data on par with the human brain.

## 2.3 Data Scales

For analytics purposes, data is grouped into four scales: nominal, ordinal, interval, and ratio (Walliman, 2010). Understanding these scales is essential, since different analytics techniques work with different scales — sometimes requiring a technique choice, or a scale conversion, to match the intended method.

### Nominal Scale

Nominal data represents named groups or categories with no inherent order. For example:

- Gender: male, female
- Faculty: Engineering, Science, Information Technology
- Blood type: A, B, O, AB
- Appliances: TV, fan, refrigerator, computer, mobile phone
- City: Bangkok, Tokyo, London, Suphanburi, Beijing

Such values cannot be ranked — for example, in the appliance group, we cannot say a refrigerator is “greater” or “less” than a fan.

### Ordinal Scale

Ordinal data is a step above nominal data: it represents categories that can be ranked, but the difference between ranks cannot be quantified. For example:

- Grade: A, B+, B, C+, C, D+, D, F
- Shirt size: XS, S, M, L, XL
- Survey response: strongly agree, agree, neutral, disagree, strongly disagree
- Sentiment: positive, neutral, negative
- Risk level: none, low, medium, high

These values can be ordered — grade A is higher than B+ — but we cannot say the gap between A and B+ equals the gap between B+ and B.

### Interval Scale

Interval data is a step above ordinal data: the difference between values can be measured, but there is no true zero — zero is just another value, not “nothing,” so ratios between values are not meaningful. For example:

- Temperature in Celsius: ..., -2, -1, 0, 1, 2, 3, 4, 5, ...
- Temperature in Fahrenheit: ..., -2, -1, 0, 1, 2, 3, 4, 5, ...
- pH value: 1, 2, 3, ..., 14
- Calendar dates: 2021-06-11, 2021-06-14, 2021-10-15, ...

We can say 20°C is 5 degrees more than 15°C, but we cannot say 0°C means “no heat,” nor use 0°C as the same reference point as 0°F, nor say 20°C is “twice as hot” as 10°C.

### Ratio Scale

Ratio data is the highest level, one step above interval data, with a true zero — meaning zero genuinely represents “none,” providing a valid reference point across the scale, so ratios between values are meaningful. Examples include height, weight, age, monetary amount, and distance.

Because of this true zero, a distance of 0 km genuinely means “no distance,” and we can correctly say that 100 km is 10 times farther than 10 km.

## 2.4 Data Collection

Collecting data is another core task of the analytics team, since acquiring data is the essential first step of the whole process (see Figures 1.2-1 and 2.0-1): everything starts from data, followed by analysis, decision-making, and finally action. The team must therefore understand how data is obtained from different sources and the problems that can arise during collection.

### Data Sources

In statistics, data sources are grouped into primary data and secondary data:

- **Primary Data** — raw data collected directly by the analytics team from its original source (Walliman, 2010): direct surveys, self-run experiments, direct interviews, or direct sensor readings. Its advantage is accuracy and exact fit to the team's needs, at the cost of extra time and expense to acquire.
- **Secondary Data** — data the analytics team did not collect itself, but obtained from another source — an organization or individual that already gathered it, whether as raw data or pre-summarized results (Walliman, 2010): sensor data, government statistics, private-sector data, research institutes, researchers, books, or journals. Its advantages and disadvantages mirror those of primary data: it is available with little or no time and cost, but it may not fully match the analytics team's actual needs, since the team wasn't the one who framed the original data-collection goal.

For example, consider a company running a social-network application where many users post, comment, like, and share content, generating large volumes of data. This is primary data for the company; if that data (raw or summarized) is then sold or distributed to other organizations, it becomes secondary data for them.

For readers starting out in data analytics, building a primary dataset from scratch is usually impractical due to the time and cost involved, so secondary data is recommended for learning purposes. This book provides ready-to-use example datasets, described in the next section — some compiled by the author as primary data, and some drawn from popular external secondary sources. Readers can also find additional secondary data from open-data sources, which are freely accessible over the internet in a usable format, such as:

- **Kaggle** (<https://www.kaggle.com>) — a very popular hub of well-organized datasets for analytics, complete with example problems and community discussions of analysis approaches.
- **Open government data portals** from various countries, e.g., Thailand (<https://data.go.th>), the United States (<https://www.data.gov>), the United Kingdom (<https://data.gov.uk>), and Japan (<https://www.data.go.jp>).
- **DBpedia** (<https://www.dbpedia.org>) — a large Linked Open Data repository built from Wikipedia content, structured as a knowledge graph. As of 2021, DBpedia links over 4 million entities, covering places, people, works, species, organizations, and more, accessible via the SPARQL query language; readers interested in using it effectively should study the Semantic Web and SPARQL further (Heath & Bizer, 2011).

### Challenges in Data Collection

Data collection in practice is rarely straightforward. Analytics teams commonly face problems arising from the variety of data sources, structures, and recording practices (Igual & Segui, 2017), including:

- **Different file formats** — e.g., Excel, CSV, JSON, or relational databases — requiring the team to consolidate these into one ready-to-use format, typically a table or data frame.
- **Missing data** — arising from data simply not existing, not yet being known, not yet being recorded, or recording-device failure — producing null/NA values that can break analysis code; these must be handled, whether by removing or imputing values, without distorting the results.
- **Different schemas** — e.g., one dataset uses the column name “Name” while another uses “Fullname” for the same meaning, requiring renaming before the datasets can be merged; or one dataset stores full name in one column while another splits it into separate columns, requiring a consistent approach to be chosen.
- **Different data formats** — e.g., inconsistent date orderings (year-month-day, year/month/day, day/month/year, month/day/year), inconsistent month representation (numeric or abbreviated), or 2-digit vs. 4-digit years, requiring standardization, ideally to an international format.

- Different units — e.g., weight (kg, g, lb), currency (baht, yen, euro, USD), speed (mph, km/h), temperature (Celsius, Fahrenheit, Kelvin), or calendar year (Buddhist era, Common era) — requiring the team to choose and standardize on a suitable unit.
- Different frequencies — especially for time-series data, e.g., one sensor logs every minute while another logs hourly.
- Incorrect data — from recording errors or outdated records, e.g., a country's capital city that has since changed, or a customer's outdated address.

Because these challenges vary depending on how the source system (or the staff responsible for it) records data, the analytics team must recognize them, diagnose the underlying issue, and plan the initial data-processing steps needed — the subject of the next chapter.

## 2.5 Example Datasets Used in This Book

The example datasets used throughout this book are hosted at <https://github.com/Rathachai/DA-LAB>. Each is described below.

### Employee Sample Dataset

A simulated employee dataset for a fictitious company.

#### Access

<https://rathachai.github.io/DA-LAB/datasets/simple-employee-db.csv>

#### Description

**Table 2.5-1: Employee dataset description**

| Column        | Description       | Data Scale   |
|---------------|-------------------|--------------|
| eid           | Employee ID       | (identifier) |
| name          | Employee name     | Nominal      |
| gender        | Gender            | Nominal      |
| department    | Department        | Nominal      |
| age           | Age (years)       | Ratio        |
| salary        | Salary (baht)     | Ratio        |
| working_years | Years of service  | Ratio        |
| birth_place   | Province of birth | Nominal      |

#### Sample Data

**Table 2.5-2: Employee data sample**

| eid  | name     | gender | department | age | salary | working_years | birth_place |
|------|----------|--------|------------|-----|--------|---------------|-------------|
| E011 | Anda     | female | developer  | 39  | 64200  | 4             | Bangkok     |
| E012 | Bordin   | male   | developer  | 25  | 48700  | 2             | Phuket      |
| E013 | Chantana | female | developer  | 29  | 45500  | 3             | Chonburi    |
| E014 | Donlaya  | female | marketing  | 39  | 72600  | 8             | Chonburi    |
| E015 | Ekkasit  | male   | marketing  | 37  | 80500  | 7             | Suphanburi  |
| E016 | Fundee   | female | support    | 35  | 56600  | 3             | Phuket      |
| E017 | Gitiwit  | male   | support    | 26  | 42400  | 2             | Suphanburi  |
| E018 | Harit    | male   | devops     | 32  | 67700  | 5             | Bangkok     |

## Simple Vehicle Classification Dataset

A simulated dataset for a simple classification task: car vs. non-car.

### Access

<https://rathachai.github.io/DA-LAB/datasets/simple-veh-class.csv>

### Description

**Table 2.5-3: Vehicle classification dataset description**

| Column    | Description                | Data Scale   |
|-----------|----------------------------|--------------|
| vid       | Vehicle ID                 | (identifier) |
| weight_kg | Vehicle weight (kg)        | Ratio        |
| height_m  | Vehicle height (m)         | Ratio        |
| n_wheels  | Number of wheels           | Ratio        |
| vtype     | Vehicle type: car or other | Nominal      |

### Sample Data

**Table 2.5-4: Vehicle classification data sample**

| vid   | weight_kg | height_m | n_wheels | vtype |
|-------|-----------|----------|----------|-------|
| v-001 | 1031      | 1.41     | 4        | car   |
| v-002 | 885       | 1.72     | 4        | car   |
| v-003 | 977       | 1.66     | 4        | car   |
| v-009 | 1196      | 1.53     | 4        | car   |
| v-010 | 149       | 1.59     | 4        | other |
| v-014 | 2434      | 4.13     | 6        | other |
| v-018 | 2320      | 1.36     | 4        | other |
| v-020 | 2785      | 1.39     | 4        | other |

## Flu Diagnosis Dataset

Laboratory-confirmed flu diagnosis results together with recorded symptoms — body temperature, headache, and nausea — for use in learning basic patient-screening models (Grzymala-Busse, 2004).

### Access

<https://rathachai.github.io/DA-LAB/datasets/flu-test.csv>

### Description

**Table 2.5-5: Flu diagnosis dataset description**

| Column   | Description              | Data Scale   |
|----------|--------------------------|--------------|
| case     | Case number              | (identifier) |
| temp     | Body temperature range   | Ordinal      |
| headache | Whether headache present | Nominal      |

| Column | Description                 | Data Scale |
|--------|-----------------------------|------------|
| nausea | Whether nausea present      | Nominal    |
| is_flu | Lab-confirmed flu diagnosis | Nominal    |

### Sample Data

**Table 2.5-6: Flu diagnosis data sample**

| case | temp      | headache | nausea | flu |
|------|-----------|----------|--------|-----|
| 1    | high      | YES      | no     | YES |
| 2    | very_high | YES      | YES    | YES |
| 3    | normal    | no       | no     | no  |
| 4    | high      | YES      | YES    | YES |
| 5    | high      | no       | YES    | no  |
| 6    | normal    | YES      | no     | no  |
| 7    | normal    | no       | YES    | no  |

### Boston Housing Price Dataset

Housing prices in Boston (target variable medv), with 13 predictor variables (Harrison & Rubinfeld, 1978). This is a widely used dataset for learning regression analysis.

#### Access

<https://rathachai.github.io/DA-LAB/datasets/boston.csv>

#### Description

**Table 2.5-7: Boston housing dataset description**

| Column  | Description                                                     | Data Scale        |
|---------|-----------------------------------------------------------------|-------------------|
| crim    | Per-capita crime rate by town                                   | Ratio             |
| zn      | Proportion of residential land zoned for large lots             | Ratio             |
| indus   | Proportion of non-retail business acres per town                | Ratio             |
| chas    | Charles River dummy variable (1 = borders river, 0 = otherwise) | Nominal (encoded) |
| nox     | Nitrogen oxide concentration (parts per 10 million)             | Ratio             |
| rm      | Average number of rooms per dwelling                            | Ratio             |
| age     | Proportion of owner-occupied units built before 1940            | Ratio             |
| dis     | Weighted distance to employment centers                         | Ratio             |
| rad     | Index of accessibility to radial highways                       | Ratio             |
| tax     | Property tax rate per \$10,000                                  | Ratio             |
| ptratio | Pupil-teacher ratio by town                                     | Ratio             |
| black   | Proportion of Black residents by town                           | Ratio             |
| lstat   | Percentage of lower-status population                           | Ratio             |
| medv    | Median home value (in \$1,000s)                                 | Ratio             |



**Table 2.5-8: Boston housing data sample**

| crim    | zn   | indus | chas | nox   | rm    | age  | dis  | rad | ... | lstat | medv |
|---------|------|-------|------|-------|-------|------|------|-----|-----|-------|------|
| 0.00632 | 18.0 | 2.31  | 0    | 0.538 | 6.575 | 65.2 | 4.09 | 1   |     | 4.98  | 24.0 |
| 0.02731 | 0.0  | 7.07  | 0    | 0.469 | 6.421 | 78.9 | 4.97 | 2   |     | 9.14  | 21.6 |
| 0.02729 | 0.0  | 7.07  | 0    | 0.469 | 7.185 | 61.1 | 4.97 | 2   |     | 4.03  | 34.7 |
| 0.03237 | 0.0  | 2.18  | 0    | 0.458 | 6.998 | 45.8 | 6.06 | 3   |     | 2.94  | 33.4 |

## 2.6 Chapter Summary

Data collection is the very first step in turning real-world phenomena into digital data ready for use, and Internet of Things technology has made sensing and recording data far more efficient. Data can be structured, such as tables (especially CSV files), or unstructured, such as images, text, and sound — the latter requiring AI-based pre-processing to become structured before analysis. This chapter also covered how data is collected and introduced the example tabular datasets used in later chapters. Importantly, it introduced four data scales — nominal, ordinal, interval, and ratio — which form the foundation for understanding data throughout the rest of the book.

## 2.7 Review Questions

1. Write a CSV file from the flu diagnosis data in Table 2.5-6.
2. Explain whether values within the same column can be of different data types.
3. Explain why unstructured data cannot be analyzed directly without pre-processing.
4. Research how humans collected data before Internet of Things technology existed.
5. Explain whether scanning a printed page of Table 2.5-6 from this book into a computer would produce structured data, and why.
6. Using the supplementary exercises at <https://github.com/Rathachai/DA-LAB/tree/gh-pages/exercises>, write a description for each of the following datasets: (a) Titanic survivors, (b) mushroom classification, (c) drug classification, (d) mall customer behavior, (e) movie ratings.

## 2.8 References

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